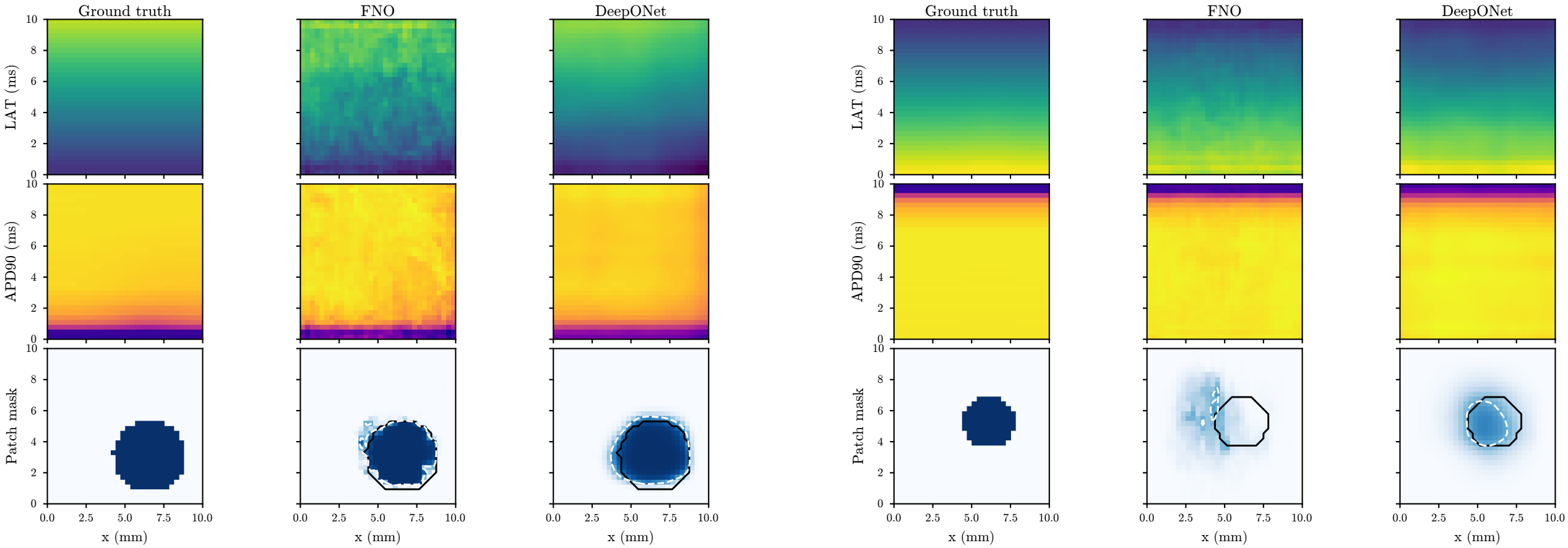
Held-out spatial examples: accurate fields can coexist with weak localization

A Common localization success
APD perturbation; FNO/Deep Dice 0.91/0.88; LAT R^2 0.90/0.95

B Largest model disagreement
Conduction slowing; FNO/Deep Dice 0.00/0.75; LAT R^2 0.98/0.99

C High-field / weak-mask failure
Conduction slowing; FNO/Deep Dice 0.00/0.00; LAT R^2 0.99/0.99

D Homogeneous false positive
Homogeneous; FNO/Deep FP area 0.001/0.068



Mask panels: black solid = true support; white dashed = predicted 0.5 contour. Field and localization maps use separately validation-selected checkpoints.